

Staff Engineer, Post-Silicon Debug & Validation Engineer

Samsung Austin Semiconductor · 3900 N Capital of Texas Hwy, Austin, TX, USA

Job ID (this archive)	SC-145
Employer	Samsung Austin Semiconductor
Location(s)	3900 N Capital of Texas Hwy, Austin, TX, USA
Posted / listing date	Posted 6 Days Ago
Career stage (archive tag)	Experienced
Location group	Texas (outside DFW)
Career lane	Test, product, quality & reliability
Pay listed	\$151,000 - \$226,600
Employer requisition	R120417
Source page	https://sec.wd3.myworkdayjobs.com/Samsung_Careers/job/3900-N-Capital-of-Texas-Hwy-Austin-TX-USA/Staff-Engineer--Post-Silicon-Debug---Validation-Engineer_R120417
Accessed	2026-09-26

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Job description

Position Summary

Samsung, a world leader in advanced semiconductor technology, is founded on a simple philosophy – the endless pursuit of excellence will create a better world for all. At Samsung Austin Research and Development Center (SARC) and Advanced Computing Lab (ACL), we are building a center of excellence for Intellectual Property (IP) that is applied to high-performance computing devices (mobile, automotive, and other custom market segments) consumed by millions of people around the world. Come build with us!

Role and Responsibilities

As a Staff GPU Post-Silicon Debug and Validation Engineer, you will contribute to the bring-up, debug, and validation of next-generation GPU and system-level IP across silicon and emulation platforms. Your work will directly impact product quality, performance, and time to market for Samsung's GPU technologies deployed across mobile and adjacent SoCs.

In this individual contributor role, you will work across silicon debug, system validation, and emulation environments to identify, analyze, and resolve complex functional and performance issues. You bring curiosity to collaborate closely with architecture, design, verification, and software teams to drive root cause analysis, improve debug efficiency, and enable robust validation strategies from bring-up through production.

- Support GPU silicon debug, design-for-debug architecture, and diagnostic development under the guidance of senior engineers
- Assist with automated SOC-level system test generation for GPU and Coherent Interconnect
- Participate in board-level hardware and software bring-up of GPU and SOC (Memory Controllers, Chip Interconnect/Fabric)
- Perform functional and parametric analysis of SOC and IP within an SOC
- Help debug and root-cause improper or inconsistent operation
- Develop software tools or scripts to support debug and test simplification and automation, leveraging AI/ML tools to accelerate analysis and automate repetitive debug workflows
- Learn and apply debug features to accelerate testing and improve test effectiveness
- Contribute to IP-level functional validation test plans
- Support SOC emulation: modify hardware, software, and test benches, run emulation, and analyze SOC behavior
- Provide debug and testing input for feature improvements to next-generation designs

Skills and Qualifications

- 6+ years of experience with a Bachelor's Degree in Computer Science/Engineering, or 4+ years of experience with a Master's Degree, or 2+ years of experience with a Ph.D.
- BS in Electrical Engineering, Computer Engineering, or comparable degree (recent or upcoming graduate); internship or academic project experience in GPU or post-silicon debug/validation is a plus
- Must have hands-on experience with AI/ML tools or techniques (e.g., using AI coding assistants, applying ML/data analysis to debug or test automation, or coursework/projects involving AI/ML) — this is a required skill for the role
- Coursework or project experience with system/board-level bring-up
- Exposure to GPU or microprocessor debug using Logic Analyzers, Scopes, or other test equipment
- Familiarity with debuggers such as Lauterbach (coursework or lab experience acceptable)
- Basic understanding of performance and power measurement techniques
- Exposure to schain, array dump, or other HW debug techniques (academic or internship experience acceptable)
- Familiarity with waveform analysis and emulator signal extraction concepts
- Exposure to IPXACT design documents or parser development (a plus)
- Foundational GPU architectural knowledge (coursework or personal projects)
- Familiarity with GPU APIs (OpenGL, OpenGL ES, Vulkan) and GPU software stacks
- Proficiency in C/C++/Object-Oriented Programming, Python, and Shell Scripting
- Basic understanding of GPU/microprocessor product and test engineering concepts

- Exposure to Linux and/or Android kernel programming
- Familiarity with BIOS/firmware programming concepts (a plus)
- Coursework or project experience with Verilog and/or UVM

Our Team

The Central Engineering organization at SARC/ACL drives technology enablement for GPU and SoC development by delivering critical engineering infrastructure, IP, and services across emulation, compute, silicon validation, and manufacturing readiness. Spanning the full development lifecycle, our teams partner closely with design and manufacturing stakeholders to support early silicon engagement, system bring-up, validation, power and performance analysis, debug, and test coverage—ensuring efficient execution and production readiness at scale.

Within Central Engineering, the Silicon & Emulation team focuses on emulation platforms, silicon validation, and system-level analysis to enable functional correctness and optimized power and performance. You will join a team that operates at the intersection of multiple engineering domains to solve real-world challenges, see the impact of your work, and experience innovation in a collaborative environment that values ownership, continuous learning, and technical excellence.

Total Rewards

At Samsung – SARC/ACL, base pay is one part of our total compensation package and is determined within a range. This provides the opportunity to progress as you grow and develop within a role. The base pay range for this role is between \$151,000 and \$226,600. Your actual base pay will depend on variables that may include your education skills, qualifications, experience, and work location.

Samsung employees have access to benefits including: medical, dental, vision, life insurance, 401(k), onsite lunch, employee purchase program, tuition assistance (after 6 months), paid time off, student loan program, wellness incentives, and many more. In addition, regular full-time employees (salaried or hourly) are eligible for MBO bonus compensation, based on company, division, and individual performance.

Additionally, this role might be eligible to participate in long term incentive plan and relocation.

This is an exempt position, which is not eligible for overtime pay under the Fair Labor Standards Act (FLSA).

Trade Secrets

By submitting an application, you agree not to disclose to Samsung or induce or encourage Samsung to use any confidential or proprietary information (including trade secrets) belonging to a current or former employer or other entity.

U.S. Export Control

This role may require access to information subject to U.S. export control laws. Applicants must be authorized to access such information or eligible for government authorization.

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